

PRAISELIN LYDIA GLADSTON

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SOFTWARE DEVELOPER | ML ENGINEER

PROFESSIONAL SUMMARY

Performance-oriented software developer with 6+ years of experience specializing in **backend systems, compiler optimization, and cloud-scale infrastructure**. Proven success in **C++, Java, and Python development**, multithreaded systems, and containerized microservices. Passionate about **developer experience**, performance tuning, and delivering robust solutions across **SaaS and on-premises** applications. Formally trained in **AI/ML** with hands-on learning.

CORE TECHNICAL SKILLS

Languages: C, C++, Java, Python, Bash, SQL

Cloud & Data: Microsoft Azure (Blob, ADF, Cosmos DB), Apache Spark, Databricks, Delta Lake

Web & Frontend: JavaScript, HTML, CSS, Java Servlets, Flask

Systems & Backend: Columnar storage engines, Compiler-level optimization, REST APIs, Multithreading

DevOps & Tooling: Docker, Git, Bash Scripting, CI/CD, Intel VTune, gdb, Postman

AI & ML: TensorFlow, scikit-learn, Numpy, Pandas, NLP, Time-Series Analysis

Visualization & Monitoring: Power BI, Grafana, custom Bash-based health checks

Practices: Agile, Test-Driven Development, Benchmarking, Code Reviews

PROFESSIONAL EXPERIENCE

Software Developer | Zoho Corporation Pvt. Ltd. – Chennai, India

May 2017 – November 2023

- Engineered a custom **C++ vectorized columnar storage engine**, increasing query performance by 40% through I/O and SIMD optimization.
- Developed **Java**-based utilities for real-time data migration with **zero downtime**, supporting high availability services.
- Built and deployed microservices with **Java Servlets** and frontend layers, enhancing developer modularity and user experience.
- Refactored **compiler subsystems** to improve **SIMD** alignment, reduce branching, and increase cache predictability.
- Delivered **Dockerized** services with automated **CI/CD pipelines**, enabling rapid deployment across staging and production.
- Performed feasibility studies for CPU-accelerated query execution, contributing to performance **research efforts**.
- Led system **profiling and monitoring** using Bash, Intel VTune, and custom logging solutions.
- Mentored junior engineers** and standardized best practices for performance tuning and maintainable code.
- Achieved over 95% **code coverage** using Jacoco for Java codes and gcov for C++.
- Effectively collaborated with internal teams**, fostering cross-functional communication and delivering win-win solutions
- Continuously tackled complex engineering challenges**, demonstrating high curiosity, initiative, and ownership in fast-paced environments.

EDUCATION AND CERTIFICATIONS

Southern Alberta Institute of Technology (SAIT) - Calgary, AB, Canada

January 2025 – August 2025

Professional Certificate: Data Science

Conestoga College – Waterloo, ON, Canada

January 2024 – August 2024

Graduate Certificate: Applied Artificial Intelligence and Machine Learning

GPA: 3.84 / 4.0

Velammal Institute of Technology (Affiliated to Anna University) – Chennai, India

June 2013 – April 2017

Bachelor of Engineering: Computer Science

CGPA: 8.7/ 10.0

RECENT PROJECTS

Skin Disease Detector and Tracker

- Built a **TensorFlow-based CNN** with Flask API and Docker containerization for real-time inference.
- Enabled disease progression tracking using time-series visualizations.

Tech: Python, TensorFlow, Flask, Docker

Sentiment Analysis

- Developed an NLP system using **scikit-learn + Reddit API (PRAW)** to perform sentiment analysis with live data feeds.
- Integrated a **Tkinter-based GUI**, from text preprocessing to model deployment.

Tech: Python, scikit-learn, SVC, TfidfVectorizer, SentimentIntensityAnalyzer

Research on LLMs (Ongoing)

- Exploring architecture and functioning of **transformers and LLMs** including layer structures, embeddings, and inference mechanisms.